

Understanding Growth & Customer Retention in TheLook eCommerce

By : Eva Arliana

Business Overview

TheLook is a fictional eCommerce platform that sells clothing products.

The dataset includes customer data, product details, order transactions, inventory, and website activity.

It is designed to simulate real-world business scenarios such as customer behavior, product performance, and operational efficiency





Business Context

**As an eCommerce business,
TheLook needs to :**

- Monitor inventory growth to avoid overstock or stockouts.
- Understand customer retention behavior
- Evaluate whether users are coming back after their first purchase
- Support decision-making in marketing and product strategy



Business Problem

The business wants to understand :

- How inventory is growing over time across product categories
- Whether customers are returning after their first purchase
- Which areas need improvement to sustain long-term growth



Objective

This analysis aims to :

- Measure monthly inventory growth (%) by product category
- Identify trends or anomalies in inventory movement
- Build monthly retention cohorts for 2022
- Evaluate customer loyalty and repeat purchase behavior

Disclaimer

- This dataset is synthetic (not real-world data).
- Results are used for analytical practice only.
- Some patterns may not reflect real business conditions.
- Assumptions may be applied where data is incomplete.



QUESTION 1 — INVENTORY GROWTH

To support the business, find monthly growth of inventory in percentage breakdown by product categories, ordered by time descendingly.

SQL Syntax – Question 1

WITH

```
monthly_inventory AS (  
    SELECT  
        DATE_TRUNC(created_at, MONTH) AS month  
        , product_category  
        , COUNT(*) AS total_inventory  
    FROM fsda-sql-01.TheLook_Ecommerce.inventory_items  
    GROUP BY 1, 2  
) ,  
  
growth_calc AS (  
    SELECT  
        month  
        , product_category  
        , total_inventory  
        , LAG(total_inventory) OVER (  
            PARTITION BY product_category  
            ORDER BY month  
        ) AS prev_inventory  
    FROM monthly_inventory  
)
```

SELECT

```
month  
 , product_category  
 , total_inventory  
 , prev_inventory  
 , CONCAT(  
     ROUND(  
         SAFE_DIVIDE(  
             total_inventory - prev_inventory  
             , prev_inventory  
         ) * 100,  
         2  
     ) ,  
     '%'  
 ) AS growth_percentage  
FROM growth_calc  
ORDER BY 1 DESC, 2;
```


Result – Question 1

Query results

[+ Create conversation](#)

Job information	Results	Visualization	JSON	Execution details	Execution graph
Row	month	product_category	total_inventory	prev_inventory	growth_percentage
1	2023-04-01 00:00:00 UTC	Accessories	368	1085	-66.08%
2	2023-04-01 00:00:00 UTC	Active	340	927	-63.32%
3	2023-04-01 00:00:00 UTC	Blazers & Jackets	103	328	-68.6%
4	2023-04-01 00:00:00 UTC	Clothing Sets	8	18	-55.56%
5	2023-04-01 00:00:00 UTC	Dresses	219	589	-62.82%
6	2023-04-01 00:00:00 UTC	Fashion Hoodies & Sweatshirts	383	1232	-68.91%
7	2023-04-01 00:00:00 UTC	Intimates	494	1349	-63.38%
8	2023-04-01 00:00:00 UTC	Jeans	470	1362	-65.49%
9	2023-04-01 00:00:00 UTC	Jumpsuits & Rompers	26	122	-78.69%
10	2023-04-01 00:00:00 UTC	Leggings	119	313	-61.98%
11	2023-04-01 00:00:00 UTC	Maternity	182	520	-65%
12	2023-04-01 00:00:00 UTC	Outerwear & Coats	344	988	-65.18%
13	2023-04-01 00:00:00 UTC	Pants	220	767	-69.84%

Query results

[+ Create conversation](#)

Job information	Results	Visualization	JSON	Execution details	Execution graph
Row	month	product_category	total_inventory	prev_inventory	growth_percentage
61	2023-02-01 00:00:00 UTC	Jumpsuits & Rompers	82	92	-10.87%
62	2023-02-01 00:00:00 UTC	Leggings	362	341	6.16%
63	2023-02-01 00:00:00 UTC	Maternity	556	564	-1.42%
64	2023-02-01 00:00:00 UTC	Outerwear & Coats	1056	993	6.34%
65	2023-02-01 00:00:00 UTC	Pants	763	725	5.24%
66	2023-02-01 00:00:00 UTC	Pants & Capris	373	376	-0.8%
67	2023-02-01 00:00:00 UTC	Plus	474	480	-1.25%
68	2023-02-01 00:00:00 UTC	Shorts	1239	1247	-0.64%
69	2023-02-01 00:00:00 UTC	Skirts	222	211	5.21%
70	2023-02-01 00:00:00 UTC	Sleep & Lounge	1329	1211	9.74%
71	2023-02-01 00:00:00 UTC	Socks	729	677	7.68%
72	2023-02-01 00:00:00 UTC	Socks & Hosiery	405	379	6.86%
73	2023-02-01 00:00:00 UTC	Suits	123	121	1.65%

Insight

Inventory growth shows both positive and negative changes across categories and months, indicating fluctuating inventory movement rather than consistent growth.

Some categories perform better than others, indicating differences in demand, inventory planning, or product lifecycle.

The inventory trend is not consistently declining. Instead, it fluctuates over time, with some categories growing while others decline. However, there are certain periods where multiple categories experience significant drops simultaneously.

Recommendation

1. Apply different inventory strategies per category based on their growth patterns.

2. Investigate whether changes are driven by demand (sales) or supply (restocking delays).

3. Monitor inventory trends regularly to quickly detect unusual drops or spikes.

4. Categories with high fluctuation should be prioritized for better planning and forecasting.

Inventory growth is dynamic and varies across categories. While some categories show growth, others experience sharp declines in certain periods, indicating the need for more adaptive and data-driven inventory management.

QUESTION 2 — RETENTION COHORT

Create monthly retention cohorts (the groups, or cohorts, can be defined based upon the date that a user completely purchased a product) and then how many of them (%) coming back for the following months in 2022.

SQL Syntax – Question 2

WITH

```
first_purchase AS (
    SELECT
        user_id
        , DATE_TRUNC(MIN(created_at), MONTH) AS cohort_month
    FROM fsda-sql-01.TheLook_Ecommerce.orders
    GROUP BY 1
),

user_activity AS (
    SELECT
        o.user_id
        , DATE_TRUNC(o.created_at, MONTH) AS order_month
        , f.cohort_month
    FROM fsda-sql-01.TheLook_Ecommerce.orders o
    JOIN first_purchase f
        ON o.user_id = f.user_id
),

cohort_index AS (
    SELECT
        user_id
        , cohort_month
        , order_month
        , DATE_DIFF(
            DATE(order_month)
            , DATE(cohort_month)
            , MONTH
        ) AS month_number
    FROM user_activity
    WHERE EXTRACT(YEAR FROM order_month) = 2022
),
```

```
cohort_data AS (
    SELECT
        cohort_month
        , month_number
        , COUNT(DISTINCT user_id) AS users_count
    FROM cohort_index
    GROUP BY 1, 2
),

cohort_size AS (
    SELECT
        cohort_month
        , COUNT(DISTINCT user_id) AS cohort_size
    FROM first_purchase
    GROUP BY 1
)

SELECT
    c.cohort_month
    , c.month_number
    , c.users_count
    , s.cohort_size
    , ROUND(
        SAFE_DIVIDE(c.users_count, s.cohort_size) * 100,
        2
    ) AS retention_rate
FROM cohort_data c
JOIN cohort_size s
    ON c.cohort_month = s.cohort_month
ORDER BY 1,2;
```

Result – Question 2

Query results						+ Create
Job information		Results	Visualization	JSON	Execution details	Execution graph
Row	cohort_month ▼	month_number ▼	users_count ▼	cohort_size ▼	retention_rate ▼	
1	2019-01-01 00:00:00 UTC	38	1	20	5.0	
2	2019-01-01 00:00:00 UTC	39	1	20	5.0	
3	2019-01-01 00:00:00 UTC	40	1	20	5.0	
4	2019-02-01 00:00:00 UTC	37	1	62	1.61	
5	2019-02-01 00:00:00 UTC	38	1	62	1.61	
6	2019-02-01 00:00:00 UTC	40	2	62	3.23	
7	2019-02-01 00:00:00 UTC	41	1	62	1.61	
8	2019-02-01 00:00:00 UTC	42	3	62	4.84	
9	2019-02-01 00:00:00 UTC	43	1	62	1.61	
10	2019-02-01 00:00:00 UTC	44	1	62	1.61	
11	2019-02-01 00:00:00 UTC	45	2	62	3.23	
12	2019-03-01 00:00:00 UTC	34	4	108	3.7	
13	2019-03-01 00:00:00 UTC	37	1	108	0.93	

Query results						+ Create
Job information		Results	Visualization	JSON	Execution details	Execution graph
Row	cohort_month ▼	month_number ▼	users_count ▼	cohort_size ▼	retention_rate ▼	
480	2022-08-01 00:00:00 UTC	2	154	2687	5.73	
481	2022-08-01 00:00:00 UTC	3	144	2687	5.36	
482	2022-08-01 00:00:00 UTC	4	169	2687	6.29	
483	2022-09-01 00:00:00 UTC	0	2745	2745	100.0	
484	2022-09-01 00:00:00 UTC	1	174	2745	6.34	
485	2022-09-01 00:00:00 UTC	2	176	2745	6.41	
486	2022-09-01 00:00:00 UTC	3	168	2745	6.12	
487	2022-10-01 00:00:00 UTC	0	3040	3040	100.0	
488	2022-10-01 00:00:00 UTC	1	225	3040	7.4	
489	2022-10-01 00:00:00 UTC	2	232	3040	7.63	
490	2022-11-01 00:00:00 UTC	0	3137	3137	100.0	
491	2022-11-01 00:00:00 UTC	1	245	3137	7.81	
492	2022-12-01 00:00:00 UTC	0	3402	3402	100.0	

Insight & Recommendation

Insight :

Customer retention drops significantly after the first purchase, decreasing from 100% to around 1-7% in the following months. This indicates that most users do not return, and overall long-term engagement remains low and inconsistent across cohorts.

Recommendation :

The business should focus on increasing repeat purchases by improving early-stage retention through post-purchase incentives, personalized marketing, and retargeting strategies, while also leveraging successful cohorts to optimize future campaigns.

Summary

This analysis highlights two key aspects of TheLook's business : **inventory movement** and **customer retention**.

Inventory trends show **fluctuating growth across product categories**, indicating the need for better planning and monitoring.

Meanwhile, cohort analysis reveals a **significant drop in customer retention after the first purchase**, with only around **1-7% of users returning** in the following months.

Overall, while the business continues to **acquire new customers**, maintaining long-term engagement remains a major challenge, emphasizing the importance of improving **retention strategies** to drive **sustainable growth**.

Thank You

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